Design and Implementation of a Computerized Library Management System for Holy Redeemer School of Cabuyao

A proposal submitted in partial fulfillment of the requirements for the Degree of Bachelor of Science in Computer Science

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CHAPTER 1

INTRODUCTION

This document showcases the System Specification for the proposed Library Management System of Holy Redeemer School of Cabuyao. a private institution located in Cabuyao, Calamba. The aim is to provide an automated solution using the school's RFID system to integrate the manual library system of the school. THis system will effectively be a faster, more accurate and safer borrowing system of the library system and student account activities, replacing the school's current manual system with a centralized digital platform.

Background of the Study

This study focuses on the development of a Library Management System for Holy Redeemer School of Cabuyao, a basic education institution that. The school is supported by a workforce composed of teaching staff, security personnel, and maintenance employees, all of whom contribute to the institution's daily operations and educational mission.

At present, the library's operations are predominantly manual. The librarian maintains records of books and other materials using Microsoft Excel, while students are issued individual index cards to document their borrowing and returning activities. Although this method has served its purpose, it has become increasingly inefficient and prone to human error as the student population and volume of library materials continue to grow. Common issues include difficulties in tracking borrowed and returned books, misplaced or damaged index cards, and discrepancies in inventory records.

In response to these challenges, the school administration has expressed the need for a computerized library management system that will automate key library processes. The proposed system aims to efficiently manage the institution's library inventory, simplify borrowing and returning procedures, and provide the librarian or administrator with a centralized database for monitoring book availability and student borrowing records. Each book will be assigned a unique QR code to accurately track which specific copy has been borrowed by each student, thereby improving accountability and record accuracy.

Statement of the Problem

The library of **Holy Redeemer School of Cabuyao** currently operates using a manual system for managing book inventories and student borrowing records. The librarian records book information through Microsoft Excel spreadsheets, while each student maintains an individual index card for tracking their borrowed materials. Although this method allows for basic record keeping, it presents significant limitations as the student population and library collection continue to grow. The manual process often leads to misplaced records, inaccurate data, and delayed book transactions. Furthermore, the absence of a centralized database makes it difficult for the librarian to monitor borrowing activities, verify book availability, and produce comprehensive reports efficiently.

General Problem

The general problem of this study is the inefficiency and inaccuracy of the current manual library management process used by **Holy Redeemer School of Cabuyao**. The existing system lacks

automation, real-time record tracking, and centralized data management, which hinder the librarian's ability to maintain accurate, organized, and readily accessible information regarding library resources and student transactions. Therefore, this study seeks to determine how a computerized library management system can be developed to address these limitations and improve the overall efficiency of library operations.

Specific Problems

1. How can the proposed system automate the recording and monitoring of borrowed and returned books to minimize human error and duplication of records?

2. How can the system provide the librarian with immediate access to student borrowing histories and book availability status?

3. How can the system uniquely identify and track each individual book copy borrowed by a student using QR code technology?

4. How can the system generate accurate and organized reports on library transactions, inventory levels, and usage statistics?

5. How can the system ensure data security, reliability, and ease of use for the librarian and administrative staff?

Objective of the Study

General Objective

The main objective of this study is to design and develop a computerized Library Management System for Holy Redeemer School of Cabuyao that will enhance the efficiency, accuracy, and accessibility of library operations. The system aims to automate key processes such as recording book transactions, managing inventory, and tracking student borrowing activities through the use of QR code technology for precise identification of individual book copies. This will enable the librarian to easily monitor which specific book was borrowed by each student and ensure accurate record keeping.

Specific Objectives

Specifically, this study aims to:

1. Develop an automated system that will record and monitor the borrowing and returning of books in real time using QR code identification.

2. Provide a centralized database that will allow the librarian to easily access, update, and manage book and student records.

3. Generate organized reports detailing book availability, borrowing history, and inventory status to support decision-making and auditing.

4. Implement QR codes for each individual book to accurately track which copy was borrowed and by which student.

5. Incorporate user authentication and data protection features to ensure system security and data integrity
6. Design an intuitive and user-friendly interface to simplify library operations for librarians and administrative staff

Analysis of the Objectives

Criteria	Justifications
Specific?	YES. The objectives clearly identify the development of a computerized library management system with defined functions such as real-time book tracking, QR code integration, centralized record management, and report generation. Each goal is precise and directly addresses the current issues of the manual system.
Measurable?	YES. The effectiveness of the system can be measured through quantifiable improvements — including reduced transaction time, minimized record discrepancies, faster retrieval of borrowing data, and accurate tracking of individual book copies via QR codes.
Attainable?	YES. The objectives are achievable using available software tools, database technologies, and QR code generation and scanning solutions. The scope of development is well within the technical capabilities and resources of the project team.
Realistic?	YES. The proposed features, such as QR-based identification, automated recording, and centralized reporting, are practical and suitable for the operational needs of the school's library. These functionalities can be implemented without requiring major infrastructure changes.
Time-bounded?	YES. The system is designed to be implementable within the timeframe of 3 months or 13-14 weeks.

Table 1.0 Analysis of the Objectives

System Statement of Scope

The Library Management System is a centralized platform designed to automate and streamline the library operations of Holy Redeemer School of Cabuyao. It replaces the current manual methods with a digital solution focused on efficiency, accuracy, and ease of use. The system's functionality is best understood through the Input-Process-Output (IPO) model.

- Input

- **Librarian/Administrator Credentials:** Username and password for secure system access.
- **Book Information:** Details such as Title, Author, ISBN, Publisher.
- **Student Information:** Details such as Student ID, Name, Grade, and Section.
- **QR Code Scan:** The unique QR code of a book is scanned during borrowing and returning transactions.
- **Transaction Data:** The system captures the unique Student Identification code, separate from the Student ID in their own registered system, and the Book's QR code to initiate a borrowing or returning process.
- **Report Queries:** User requests to generate specific reports (e.g., inventory lists, overdue books, borrowing history).

- Process
  - **User Authentication:** Validates the login credentials of the librarian or administrator. Also validates the login credentials of the Students on their own platform.
  - **Data Management:** Allows for Creating, Reading, Updating, and Deleting (CRUD) records for books and students.
  - **QR Code Generation:** Generates a unique QR code for each individual copy of a book added to the inventory.
  - **Transaction Logging:** Records borrowing events by linking a specific book (via its QR code) to a student, timestamping the transaction, and setting a due date. It also records the return of a book, updating its status to "Available."
  - **Status Tracking:** Automatically updates the status of books (e.g., from "Available" to "Borrowed" and vice versa).
  - **Report Generation:** Compiles and organizes data from the database to create reports based on user queries.
- Output
  - **Digital Inventory:** A searchable and organized list of all library books.
  - **Student Borrowing Records:** A comprehensive history of all books borrowed and returned by each student.
  - **Real-time Book Status:** On-screen information indicating whether a specific book is available, borrowed, or overdue.
  - **Generated Reports:** Printable or viewable reports, including:
    - Master list of all books.
    - List of borrowed and overdue books.
    - Transaction history for a specific student or book.
    - Overall library usage statistics.
  - **Transaction Confirmations:** On-screen notifications confirming successful borrowing or returning actions.

Major Constraints

The development of the system is subject to the several constraints that were found in the location of the school.

- Hardware Limitations: The school does have limited implementation of computer systems in their library, only having around 2 computers used by the students and a laptop that is provided to the librarian. A single computer unit would be allocated to the project.
- Data Migration: The current system that is being used by the library is dominantly manual and non automized. They only have an excel sheet as the record for their inventory of literature. We plan to use the excel sheet to automatically input all of them into the database but the school's education director did say the librarian also functions as a teacher, and thus have not finished recording all of their available books into the excel file.
- User Adaptation: The students and the librarian as well as the admin staff, will need to be taught how to handle the system at first.

Methodology

The model that will be used is Agile Development Methodology, specifically adopting the scrum framework. Agile was chosen because of its iterative and flexible approach, which allows for continuous user feedback and rapid adaptation to changing requirements and feedback. The methodology ensures that each system component is developed incrementally, tested thoroughly, and refined based on the client's feedback, particularly the librarian and then the student body.

The Agile methodology divides the project into multiple short development cycles called sprints. Each sprint produces a functional version of the system that can and will be evaluated by the client. These sprints continue to happen till the system meets all the user/client's requirements and are to their satisfaction.

The process starts with gathering and prioritizing requirements to know what the project needs and the clients require. The team then works on a selected set of tasks for each sprint, aiming to produce a working part of the system by the end of each sprint. Regular meetings are held to discuss progress, challenges, and adjustments to ensure that the project stays on track. Testing is done continuously throughout the development to identify and fix issues early on, to improve work and reduce rework on later stages.

Agile was chosen because it supports the active collaboration between developers and the clients, which the client wanted. It encourages adaptability to the feedback gathered from the client / end users testing the working product at each cycle. Its focus on teamwork, communication, and incremental progress makes it an effective and efficient approach for developing the system that the client requires.

Review of Related Literature and Studies

The development of automated library management systems represents a critical intersection of information technology and educational resource management, reflecting the broader digital transformation occurring across academic institutions worldwide. As educational environments continue to evolve in response to technological advancements and increasing information demands, libraries must

adapt their operational frameworks to meet contemporary expectations for efficiency, accessibility, and data-driven decision-making.

This section synthesizes existing literature and studies to establish the theoretical and practical foundations for implementing a Library Management System that integrates the school's existing RFID system for student identification with QR code technology for individual book copy tracking, examining the evolution of library automation from traditional manual cataloging methods to sophisticated digital platforms, technological implementations of hybrid RFID-QR code systems in library environments, and contemporary research findings that inform the proposed system's design and functionality.

This review examines how the integration of RFID and QR code-based library automation improves student authentication, individual book tracking, real-time circulation management, user-friendly transaction processing, security, and inventory management. By analyzing studies on RFID and QR code use in libraries, it identifies best practices and strategies for creating an efficient, user-centered hybrid system that leverages existing institutional infrastructure, enhances accountability, and replaces manual record-keeping in educational institutions.

Evolution of Library Management Technologies

Library automation has transformed from basic cataloging systems to broad digital platforms that integrate multiple functions including circulation, acquisitions, and user management. Research indicates that automated library systems significantly streamline processes, reduce human error, and improve user access to information resources (Bhati et al., 2024).

The transition from manual record-keeping methods to computerized systems addresses fundamental challenges in resource management, particularly in educational institutions experiencing growth in both user populations and collection volumes (Nimbhorkar, 2024). Modern Library Management Systems replace traditional manual processes with streamlined, automated frameworks that enhance data accuracy, operational efficiency, and user experience through web-based platforms offering centralized control over library resources and enabling real-time updates to book status and user interactions (Bhati et al., 2024).

Library automation eliminates manual processes, improving the accuracy and efficiency of library operations, allowing for quicker access to resources, and enhancing the overall user experience while enabling the integration of traditional and digital collections (Rajwanti, 2025, p. 112)

QR Code Technology in Library Applications

QR code technology has emerged as a promising solution for library resource management, providing an easy and convenient way for users to access library and management information services (Turamari, 2022). The implementation of QR codes in academic libraries offers several distinct advantages including open-source availability at no cost, simple implementation processes, and user-friendly technology requiring only smartphones with QR code scanning capabilities (Turamari, 2022).

Research shows that QR codes enable users to connect with library resources such as Web-OPAC, Institutional Digital Repositories, library websites, e-resources, library guides, and new arrivals without wasting users' time, while providing large storage data capacity in a small space (Turamari, 2022).

The application of QR code technology in libraries facilitates document delivery services, increases information literacy and technology literacy among library users, and effectively converts information needs from physical to virtual mode instantaneously (Turamari, 2022; Chikkamanju & Ambika, 2024).

Bhati, P., Komal, Prince, and Saxena, S. (2024) discussed how modern library systems have become more efficient with the help of web-based automation. Their study shows that by using an online platform, libraries can easily organize their materials, monitor usage, and generate real-time reports. This research connects to the Holy Redeemer project since it also aims to move away from manual record keeping toward a centralized web system. The proposed system will allow both librarians and administrators to manage book records, transactions, and user data faster and more accurately. Bhati and colleagues also highlight that web systems help in reducing errors caused by manual handling and improve accessibility for users. This is similar to the project's goal of providing a more reliable way to track books and student borrowing history. The study gives a strong foundation for creating a more connected and automated school library system. Overall, it supports the project's vision of a digital platform that improves library efficiency and organization.

Chikkamanju and Ambika (2024) explained how QR code technology can transform library operations by making them more accessible and interactive. Their research focused on how QR codes can help users easily access library information and services with just their smartphones. This directly relates to the Holy Redeemer project, which uses QR codes to identify every physical book and track it when borrowed or returned. The authors describe QR codes as a low-cost and user-friendly solution that can enhance user experience without needing complicated hardware. The system in Holy Redeemer applies these same advantages by letting librarians and students scan codes to record transactions quickly. This helps avoid human errors and improves the speed of service in the library. The study shows that even small schools can adopt this technology effectively without large financial costs. It supports the project's goal of using simple yet powerful technology to modernize the library process.

Verma and Nair (2023) wrote a case study about implementing an electronic resource management system in a large academic library. Their study shows how digital tools help librarians manage online and physical resources from one platform. This idea connects to the Holy Redeemer project because it also centralizes library operations under a single management system. Verma and Nair emphasize the value of organized metadata, which allows for quick retrieval and easy report generation. The Holy Redeemer system follows the same logic by allowing librarians to produce transaction and inventory reports directly from its database. The study also shows that a structured digital platform improves accountability and tracking accuracy. By adopting similar methods, the project ensures that every book and borrowing activity can be monitored properly. The study provides a model for how even small institutions can benefit from organized and data-driven library management.

Data Gathering Procedures and Analysis of Outputs

This section outlines the procedures and methodologies that will be followed to gather the data necessary for this research, providing a clear and systematic approach to the collecting and analyzing information the purpose of this section is to explain and justify the methods and results that will be gathered.

Data Gathering Methods

The primary method will be through observations and interviews with the staff, primarily the librarian and the administrative staff, and a number of students. The interview will be conducted with the administrative staff first, then the librarian, then the students of Holy Redeemer School of Cabuyao. The interviews will be on how they are currently using the library and the system for borrowing and returning books, as well as the recording of all of their inventory. The interviews will also tackle how the students and the staff feel about their current system, taking into account what they might want to be changed, what they want to keep and what specific aspects they want to keep. This will output transcripts of the interviews with students, the librarian, and the administrative head.

Besides interviews with the student body and the potential users, document analysis will be made to study the current database and documents that the school is currently using for their own library system. This will study the formatting, the details of which they have taken note of, making sure that the database design as well as the process which they will go through once the system is adopted, are all familiar to a degree. This will output data analysis of the system and data formatting already done by the current system.

During the interview with the clients, the teacher-librarian, Miss Agee, expressed the want of having a quicker and more organized way to the current system that they had. They have been designing and printing physical copies of their own Library cards that the librarian gives to the students, who then fill up the card and attach a 1x1 picture on it. This Library card has the history of the books that the student has borrowed with the signature of the librarian and date borrowed as well. We have also discovered that the currently assigned teacher-librarian has manually entered the masterlist of books in a physical record book that they have at the library.

FUNCTIONAL AND DATA DESCRIPTION

System architecture

The Library Management System is architected upon a robust Three-Tier Client-Server Model, which guarantees the separation of concerns, enhances system security, and supports future scalability. This architecture comprises the Presentation Layer, the Application Layer, and the Data Layer. The design mandate requires that the system seamlessly integrate the school's existing Radio-Frequency Identification (RFID) technology, used for student identification, with a newly implemented Quick Response (QR) Code system for tracking individual book assets.

Architecture Model

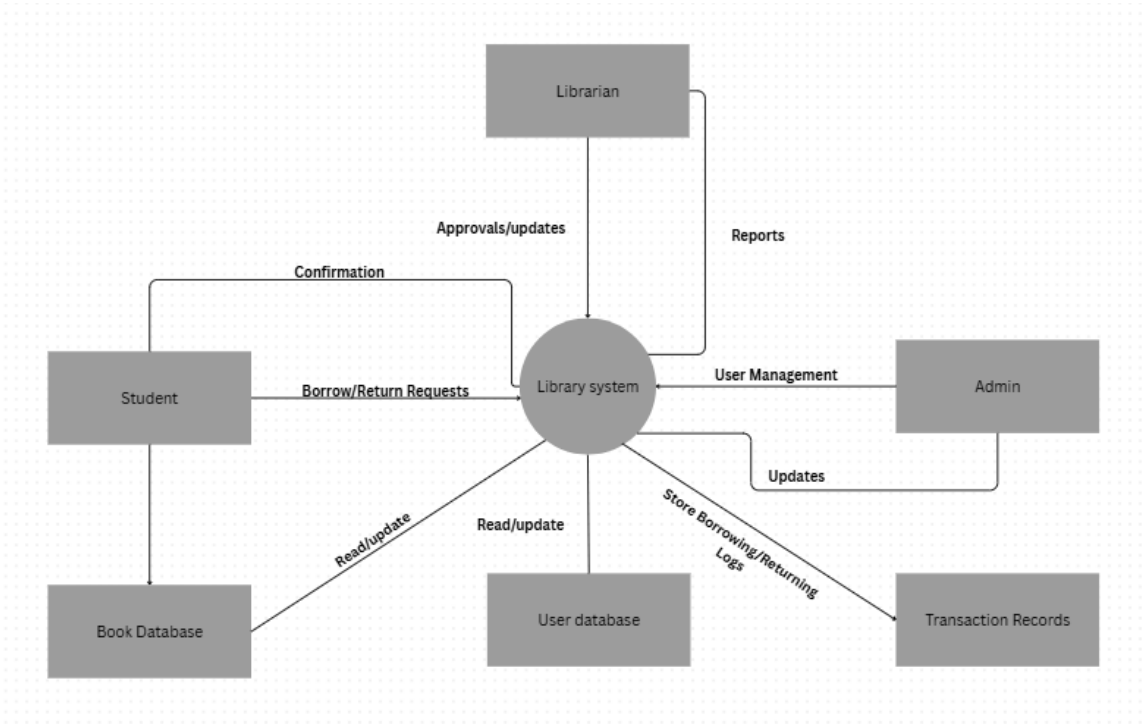


Figure 1.0 Architecture Model

The system shows the architecture and data flow of the Holy Redeemer Library system. This illustrates how the different users and databases will interact within a centralized server that manages a library system. At the center is the library system which will act as the main controller connecting all the databases such as the student, librarian, and Admin, who are the primary users who perform various tasks such as borrowing and returning , managing and generating reports. This way the architecture ensures us that every interaction are centralized and properly managed.

Subsystem Overview

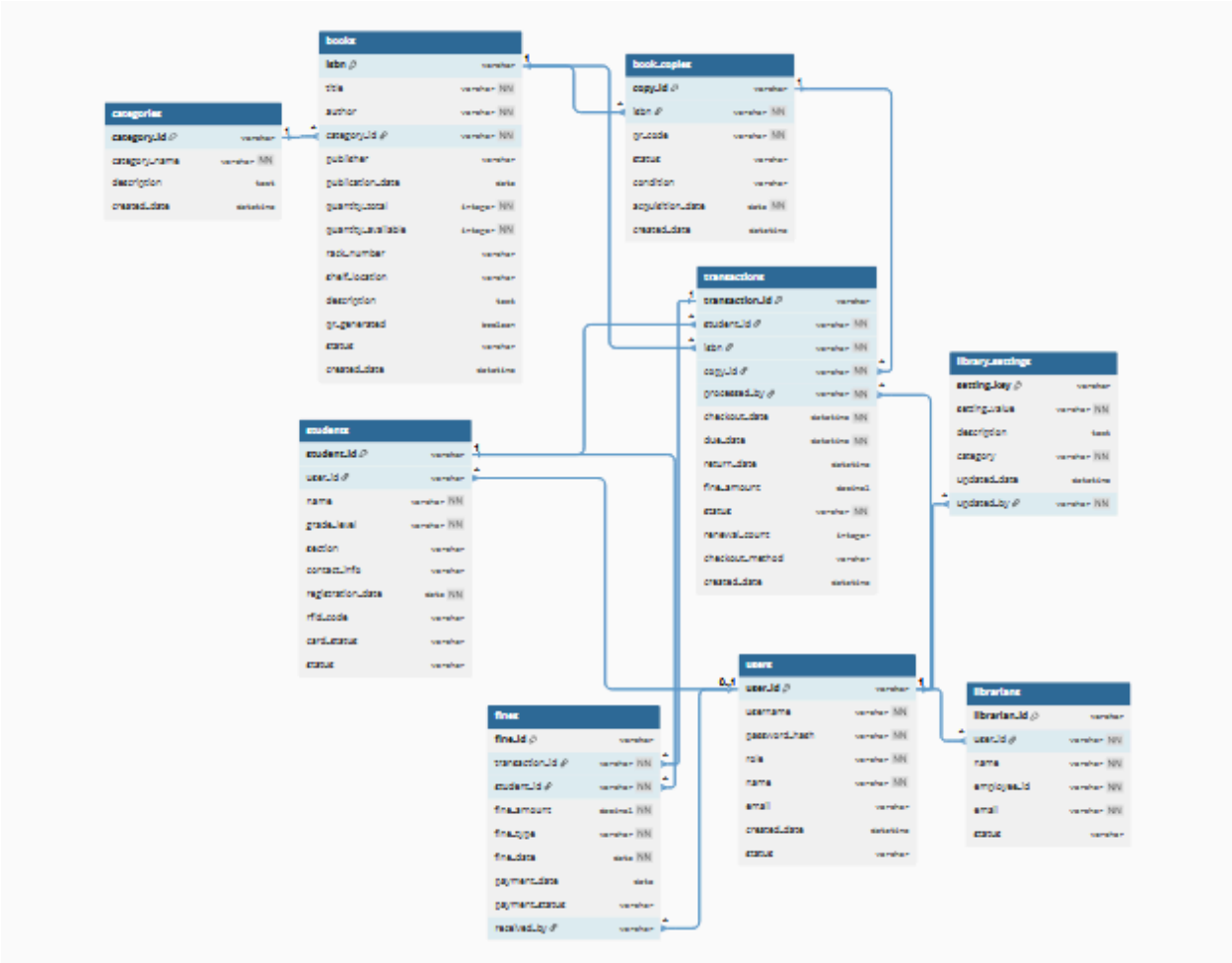
The system's boundary is defined by the Context Flow Diagram (CFD), which illustrates the core functional relationship between the central processing unit—the Library Management System—and its three external entities: Student, Librarian, and Admin. The Student is the initiating entity for transactions, providing an RFID Signal or ID Tap Data to request a loan or return, receiving a Transaction Confirmation in return. The Librarian, as the primary operator, engages in a secure HTTPS Session to provide Credentials, CRUD Data (for records), and Report Queries, receiving the Interface/Dashboard and Generated Reports. The Admin maintains the highest level of security, supplying System Configuration Parameters and User Account Provisioning data, which is crucial for managing the Librarian accounts and overseeing system

integrity. This model clarifies the system's external interfaces and the defined roles of interaction.

Data Description

Top-level data objects that will be managed/manipulated by the system or product are described in this section.

System level data model and Relationships



System level data model and Relationships

Data dictionary

A reference to the data dictionary is provided. The dictionary is maintained in electronic form.

students

Attribute	**Data Type**	**Constraint**	**Description**
Student_ID	INT	PK	The primary unique identifier for the student record.
RFID_Code	VARCHAR(15)	UNIQUE	The unique code read by the RFID scanner; used for fast, secure identification.
First_Name	VARCHAR(50)	Not Null	The student's given name.
Last_Name	VARCHAR(50)	Not Null	The student's family name.
Grade_Level	INT	Not Null	The student's current academic year level.
Section	VARCHAR(10)	Not Null	The student's assigned class

			section.
Status	VARCHAR(10)	Not Null	Current status (e.g., 'Active', 'Inactive', 'Graduated').

Book

Attribute	**Data Type**	**Constraint**	**Description**
Book_ID	VARCHAR(20)	PK	The unique identifier for this *specific physical copy* (e.g., used for inventory tags).
QRCode	VARCHAR(30)	UNIQUE	The unique QR code string assigned to this book copy for transaction scanning.
Title	VARCHAR(255)	Not Null	The full title of the book.
Author	VARCHAR(100)	Not Null	The author(s) of the book.

ISBN	VARCHAR(17)	Not Null	International Standard Book Number (unique per edition/title, not per copy).
Publisher	VARCHAR(100)	Optional	The publishing company of the book.
Current_Status	VARCHAR(15)	Not Null	Indicates availability: 'Available', 'Borrowed', 'Lost', 'Damaged'.

Transaction

Attribute	**Data Type**	**Constraint**	**Description**
Transaction_ID	INT	PK	Unique identifier for each individual loan/return event.
Student_ID	INT	FK (Student)	Links to the student who initiated the transaction.
Book_ID	VARCHAR(20)	FK (Book)	Links to the specific book copy involved

			in the transaction.
Librarian_ID	INT	FK (Librarian)	Links to the librarian who processed the transaction.
Borrow_Date	DATETIME	Not Null	Timestamp of when the book was officially checked out.
Due_Date	DATE	Not Null	The date the book is required to be returned.
Return_Date	DATETIME	Optional	The actual date/time the book was checked back in (Null if still borrowed).
Fine_Amount	DECIMAL(6,2)	Not Null	Any calculated fines incurred (defaults to 0.00).

Librarian

Attribute	**Data Type**	**Constraint**	**Description**

Librarian_ID	INT	PK	The primary unique identifier for the staff member's account.
Username	VARCHAR(50)	UNIQUE	The unique username used for system login authentication.
Password_Hash	VARCHAR(255)	Not Null	Stores the securely hashed and salted password (e.g., using Bcrypt).
Full_Name	VARCHAR(100)	Not Null	The full name of the librarian.
Contact_Email	VARCHAR(100)	Not Null	Email address for system notifications and password recovery.

Admin

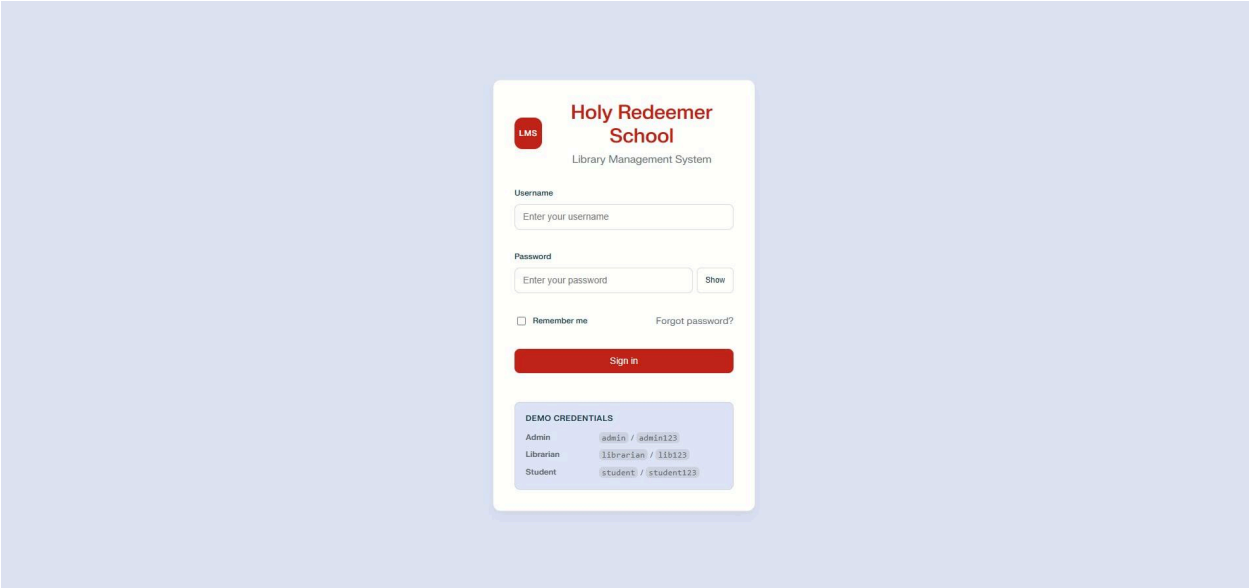
Attribute	**Data Type**	**Constraint**	**Description**
Admin_ID	INT	PK	The primary unique identifier for the administrator's

			account.
Username	VARCHAR(50)	UNIQUE	The unique username for administrative login.
Password_Hash	VARCHAR(255)	Not Null	Stores the securely hashed and salted password for system administration.
Full_Name	VARCHAR(100)	Not Null	The full name of the administrator.
Contact_Email	VARCHAR(100)	Not Null	Required email address for system alerts and security communications.

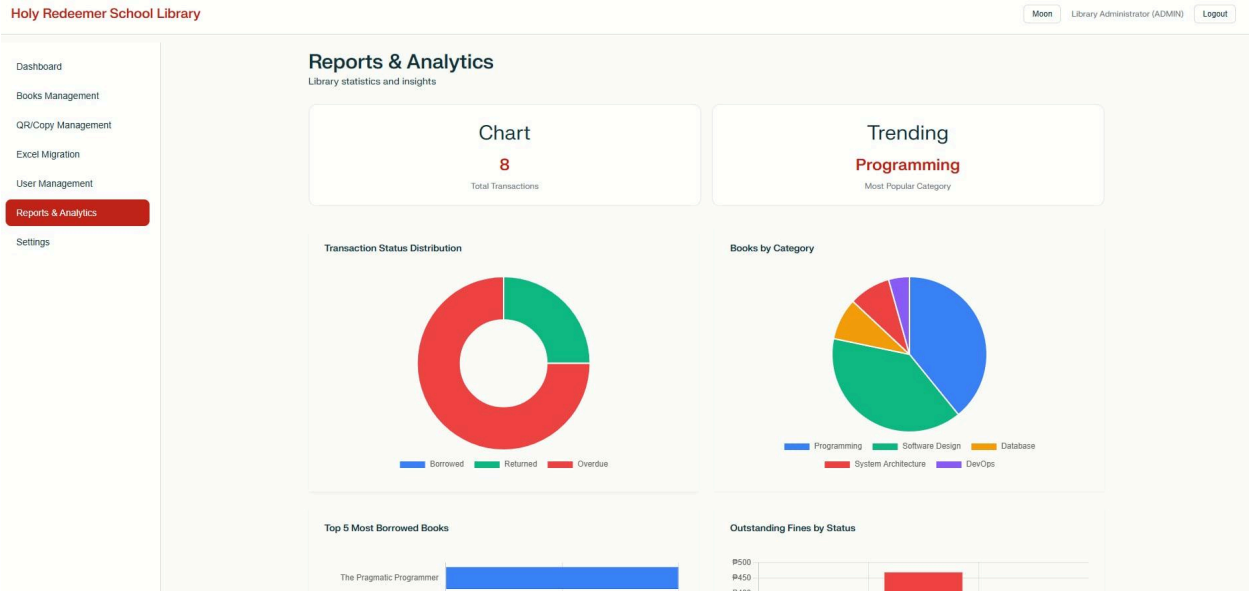
Interface Description

Graphical User interface

Login Page

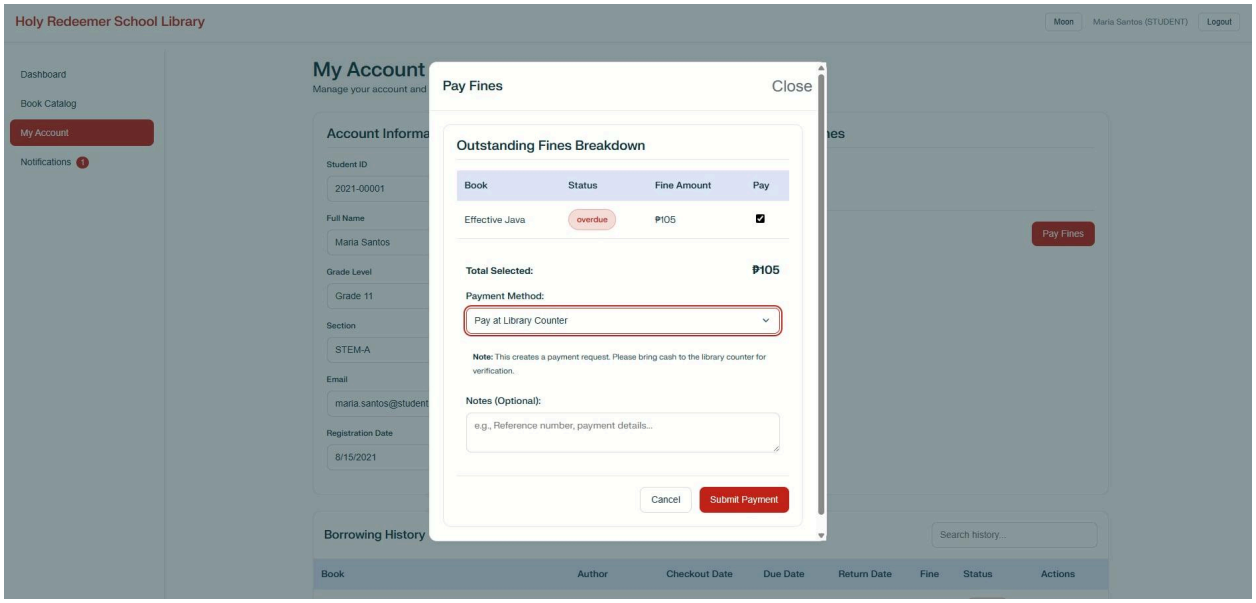


Analytics Page

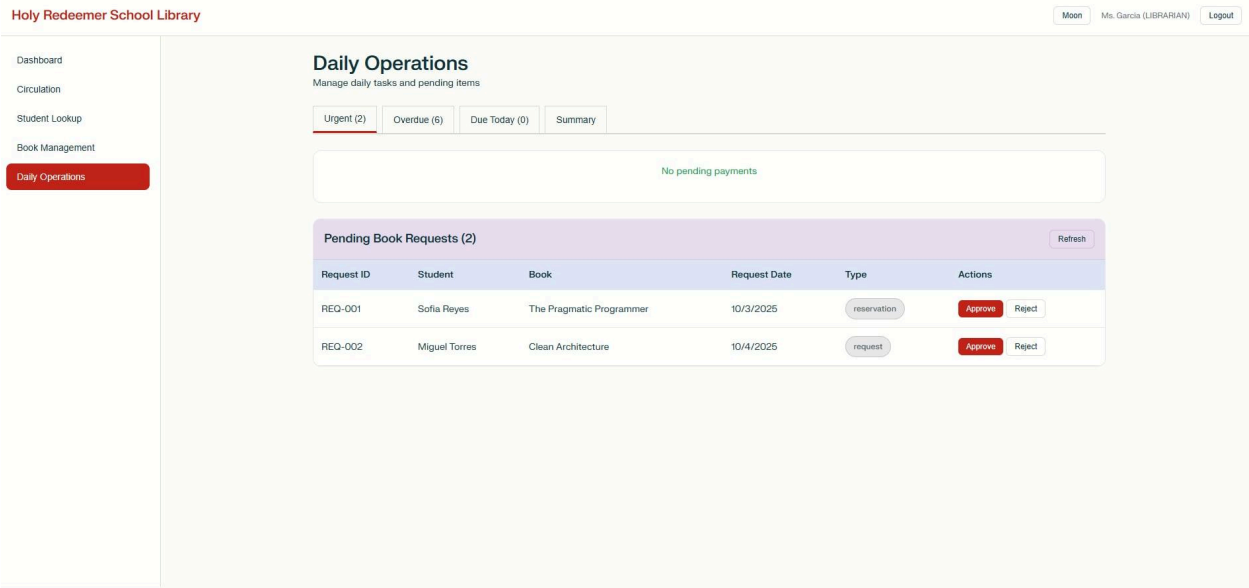


Books by QR Management Page

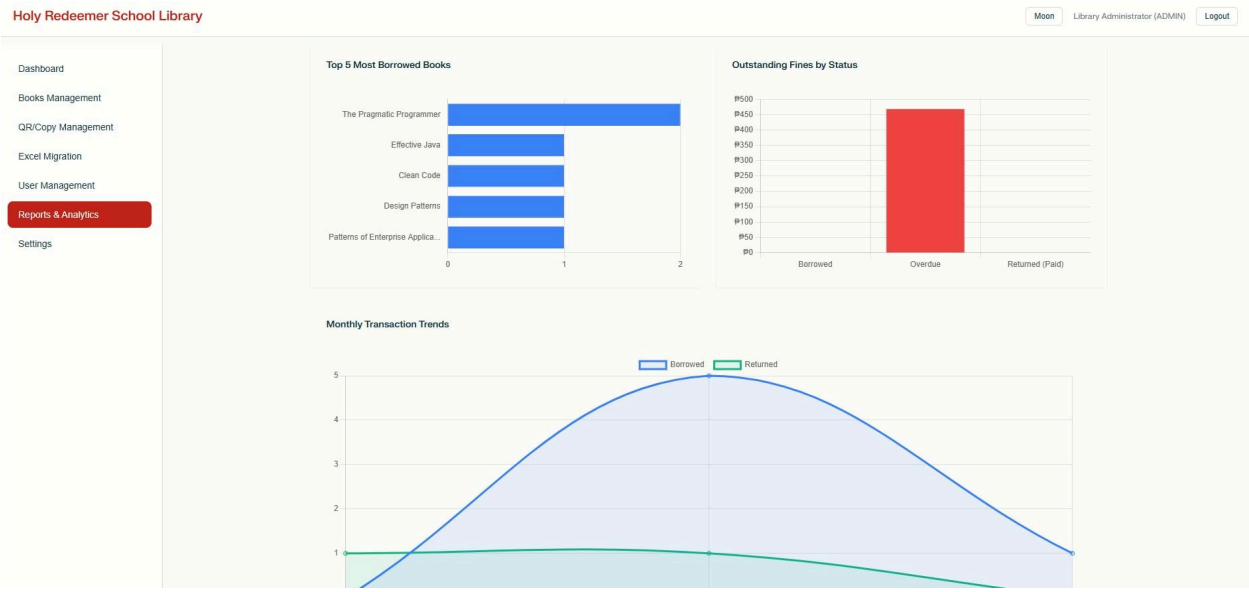
Another Page



Daily Operations Page



Analytics Page



System interfaces

Admin Features Overview

New School Year Setup
This feature allows the Library Administrator to efficiently prepare the library system for a new academic year. Through the admin dashboard, the user can review and update the current book inventory, add new materials aligned with the upcoming curriculum, and register new and transferring students. The system also enables the generation of reports summarizing the previous year's circulation data and the updating of library policies. Overall, this function ensures the library begins each school year fully equipped with an updated catalog, complete student records, and current operational policies.

Book Inventory Management
The Book Inventory Management module provides administrators with tools to maintain a complete and accurate record of all library materials. It supports adding new books with details such as ISBN, author, and category, updating records for new editions, and removing outdated or damaged books. Administrators can also organize books by category and assign shelf locations for easy tracking. The system further allows the generation of inventory reports, ensuring that the school's collection remains current, organized, and ready for use by students and staff.

Student Registration Management
This feature enables administrators to manage all student and staff library accounts. Through the User Management section, the admin can import data from the school's registration system, create individual accounts for students, and assign grade levels and sections. It also allows for handling special cases such as transferees and graduates, as well as managing librarian permissions. The outcome is a secure and organized system where every user has the correct access rights and borrowing privileges.

Monthly Library Report
The Monthly Library Report function provides administrators with insights into library performance through comprehensive analytics. It generates reports showing borrowing trends, popular book categories, overdue items, and fine collections. The system presents data visually through graphs and charts, and allows exporting reports for administrative review. This helps administrators make informed decisions, assess library engagement, and demonstrate the value of library resources to the school management.

Librarian Features Overview

Morning Opening Routine
The Morning Opening Routine equips librarians with a structured start to their workday. By logging into their dashboard, librarians can review books due for return, track overdue items, check pending reservations, and view daily statistics. This overview allows them to prioritize tasks efficiently and address maintenance issues if necessary. The feature ensures librarians begin the day prepared, organized, and aware of all ongoing library operations.

Student Book Checkout
This function streamlines the borrowing process for students. Librarians can verify student identities, check borrowing limits, and assist in searching or recommending books. The system automatically records the checkout, calculates the due date, and issues a receipt via print or email. With automated tracking and clear return procedures, the checkout process becomes faster and more accurate, improving service quality and reducing manual errors.

Research Assistance
The Research Assistance feature helps librarians support students in finding suitable learning materials for their projects. It includes tools for catalog searches, book availability checks, and placing holds for unavailable books. Librarians can suggest related resources and guide students through the borrowing process. This function promotes academic success by enhancing students' research skills and ensuring access to relevant study materials.

Fine Collection
The Fine Collection module ensures accountability for overdue books. When a student returns a book past its due date, the system automatically computes the fine. The librarian can process payments, issue receipts, and clear the student's record in real time. This promotes responsibility among borrowers while maintaining proper financial records and reinforcing library policies.

Student Features Overview:

Research Project Discovery
This feature empowers students to find and reserve materials for their academic projects. Using keyword search and category filters, students can browse available books, read summaries, and reserve items that are checked out. The system provides details such as availability and due dates, allowing students to efficiently gather resources needed for their assignments.

Weekly Reading Selection
The Weekly Reading Selection module encourages reading habits among younger students. Through the student dashboard, users can browse books by genre, age group, or category, read short descriptions, and check ratings before borrowing. The system enforces borrowing limits and sends notifications for due dates, promoting responsible reading and continuous learning.

Personal Account Management
This feature allows students to take control of their library accounts. They can view borrowed books, monitor due dates, check fines, review reading history, and update personal details. The system also supports online book renewal, preventing overdue penalties. This promotes transparency and accountability while helping students manage their academic resources efficiently.

Mobile Library Access
The Mobile Library Access function ensures that students can manage their accounts anytime, anywhere. Optimized for mobile browsers, it allows students to log in, view borrowed books, renew items, search for new materials, and receive due date

notifications directly from their phones. This flexibility makes library management more convenient for busy students.

Cross-Role Workflows Overview

New Student Orientation
This integrated process connects the roles of Admin, Librarian, and Student to onboard new users efficiently. The admin registers students in the system, the librarian introduces them to library rules and features, and the students complete their first book checkout. The process ends with a report confirming orientation completion, ensuring every new student is equipped with the knowledge to use library resources responsibly.

End-of-Semester Returns
This workflow ensures all borrowed materials are returned and accounts cleared at the end of each semester. The admin generates overdue lists, librarians send reminders, and students return books or pay fines. After processing, the admin generates a final circulation report, ensuring the library is reset and ready for the next term.

Lost Book Resolution
The Lost Book Resolution feature ensures proper handling of missing materials. When a student reports a lost book, the librarian verifies the report, and the admin determines the appropriate fee or replacement requirement. Once resolved, the librarian updates the catalog and clears the student's record. This transparent process maintains fairness and accurate inventory tracking.

Library Assessment
This workflow focuses on library improvement through collaboration between librarians and administrators. Librarians can recommend new book categories or materials based on observed student demand. Administrators then analyze data, approve purchases, and generate return-on-investment reports. This ensures data-driven decisions that enhance the library's educational value.

USAGE SCENARIO

This section provides a usage scenario for the software. It organized information collected during requirements elicitation into use-cases.

User profiles

The profiles of all user categories are described here.

Student

Attribute	Data type	Constraint	Entity	Description
Stundent_ID	Integer	Primary Key	Student	Unique student identifier
RFID_Code	Varchar	Unique	Student	Unique RFID tag assigned to each student
Book_ID	Varchar	Not Null	Book	Unique ID for each book
Status	Varchar	Not Null	Book	Indicates if book is available or borrowed

This section highlights the inner workings of the student user category of the RFID-Based library system of the Holy Redeemer School. The students are the main users who will interact with the RFID system with their ID cards. Each student has a record stored in the database and is uniquely identified using the unique Stundent_ID which serves as the primary key for the entity.

The RFID code is assigned to every student and is linked to their student ID, making it possible for fast and accurate identification during the borrowing and returning stage. The system will also associate each student with their Book_ID when borrowing library items. While the Status attribute tracks whenever the book is currently borrowed or available. This data structure ensures the students activities are properly recorded and managed in real time

The Student entity plays a vital role in maintaining accurate data of borrowing activities, ensuring availability, and promoting organized record keeping within the school's digital environment.

Librarian

Attribute	Data Type	Constraint	Entity	Description
Librarian_ID	Integer	Primary Key	Librarian	Unique identifier assigned to each librarian
Librarian_Name	Varchar	Not Null	Librarian	Full name of the librarian
Username	Varchar	Unique	Librarian	System Login username for

				the librarian
Password	Varchar	Not null	Librarian	Encrypted password for librarian login
Email	Varchar	Optional	Librarian	Email address for communication and password recovery

The library user category is responsible for managing all library related operations. Librarians will use the platform to add, update and monitor book records, generate various reports and approve or reject book borrowing and returning requests. Each librarian has a unique Librarian_ID, which acts as the primary key for identifying the records in the database.

To ensure secure access, each librarian is given a unique username and password to ensure the security of the system and its information. All this and the librarian's name will be the combination for the authentication. The Librarian_Name is the field where it will store the librarian's name, while the Email field is just there for communication purposes, notifications and password recovery.

This data structure will ensure security, accountability, and efficiency when managing the system. By maintaining the system it organizes librarian records, the system helps monitor user actions, strengthens system integrity, and ensures that only authorized personnel can modify library data.

Admin

Attribute	Data Type	Constraint	Entity	Description
Admin_ID	Integer	Primary	Admin	Unique identifier assigned to the administrator
Admin_Name	VarChar	Not Null	Admin	Full name of the system administrator
Username	VarChar	Unique	Admin	Login username for

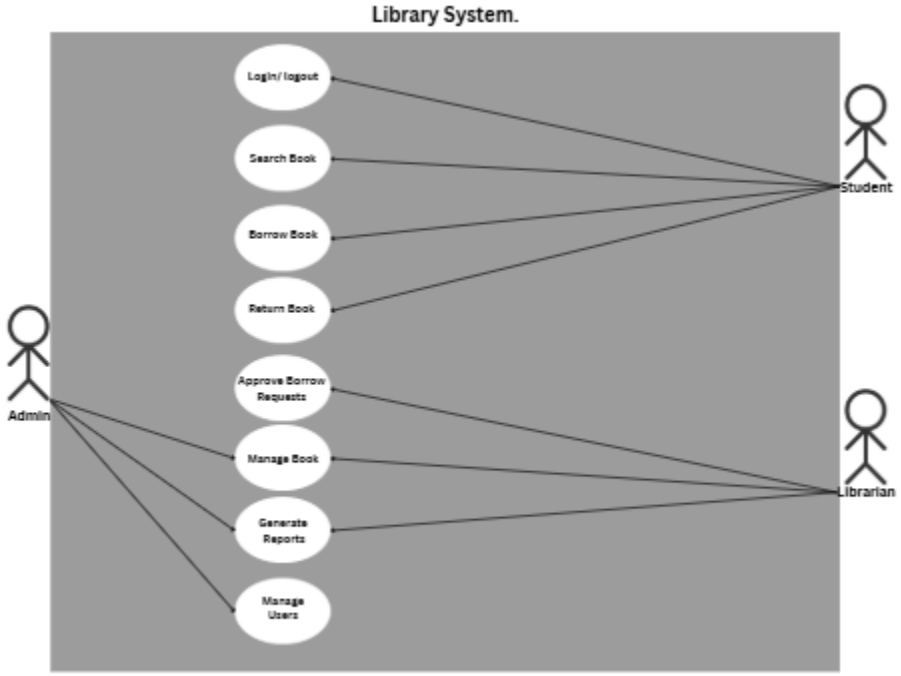
				administrative access
Password	VarChar	Not Null	Admin	Encrypted password for system authentication
Email	VarChar	Not Null	Admin	Email used for communication and account recovery

The admin user category is shown to be the highest level of system authority and has complete access to all operations and databases within the system. The administration manages all accounts, oversees all data integrity and maintains the database. The Admin_ID will serve as the primary key ensuring that all administrative accounts are unique when being identified.

The Admin_name field contains the admin's name, while the Username and Password fields are used to identify and maintain system security. The email field acts as a requirement to keep facilitating system notifications, password resets, and communication with other users. With these attributes the administrator can ensure proper configuration and maintenance of system functionalities.

Use-cases

1. Student taps id - student identifies the student account.
2. Librarian selects book - system logs borrowing
3. Admin views reports on usage, inventory, and student activity.



CHAPTER 2

INTRODUCTION

A project plan is a formal, strategic document that defines the methodology, timeline, tasks, and resource allocation required to successfully execute the system development. This plan serves as the operational blueprint for the Library Management System project, translating the functional requirements of Chapter 1 into an actionable set of developmental activities. Utilizing the Agile-Scrum framework, this chapter details the Work Breakdown Structure (WBS), the Staff Organization and responsibilities, the estimated effort required, and a proactive Risk Management strategy. By establishing a clear, iterative roadmap, the project plan ensures that the project remains within scope, is delivered on time, and meets

the quality standards required to automate and secure Holy Redeemer School of Cabuyao's library operations.

Project Scope

The digitalized library system aims to address the manual problem that the Holy Redeemer, School of Cabuyao, experiences with their completely manual system for their library. The system will leverage an online platform that the librarian/staff and the students can access to be able to see all of the inventory that the staff wants the students to see, the students and their history of what books they have borrowed, as well as keep track if the student has had enough books borrowed to satisfy their academic requirement at the end of the school year. This system was designed to automate the borrowing/returning system that they have, as well as set as the foundation for future improvements that the school is currently implementing all over their institution.

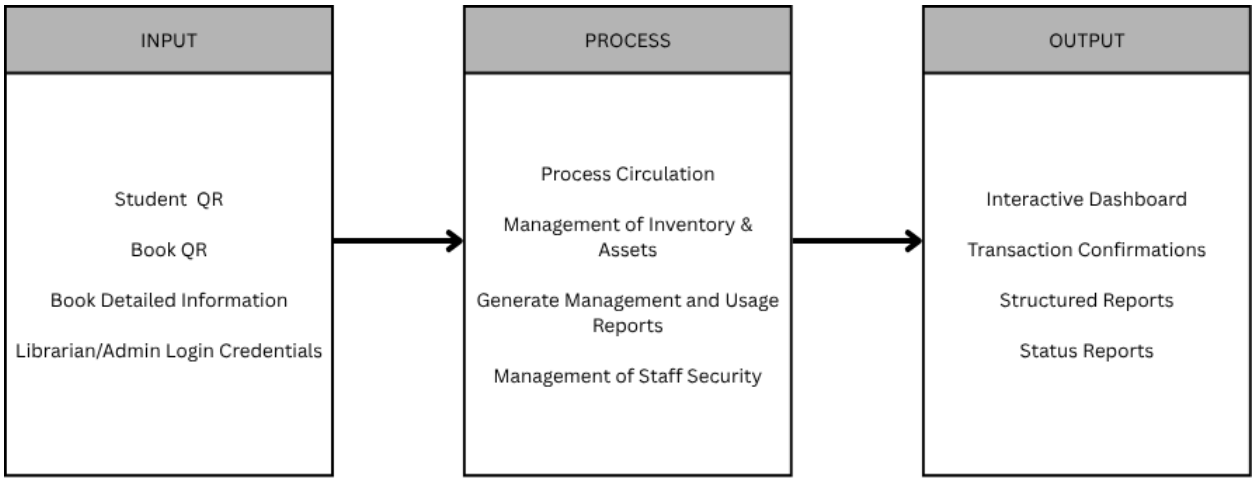


Figure X: IPO Model

Figure X presents the Input-Process-Output Model of the library system. It provides a structured overview of how the library system will function, once the system is fully developed and implemented.

In the input phase, the librarian/staff will have to control the inputs that the system will have to process. The inputs will start the most of the processes in the system. The staff/librarian will have to primarily scan the QR code of both the book and the student that will be borrowing the book to register it at the system, changing note of it in the database and updating the data where it is displayed in the system. This is also where a new record for the book is started, with the staff inputting in the details of the book, including the connected QR code.

In the Process phase, the system has multiple distinct subsystems that are crucial to meet the requirements of the client. The core system will be process circulation where the handling of the logic that presides over the borrowing and returning transactions reside. This will contain the functions that will write data on the main server. Management of Inventory and Assets execute all the CRUD operations,

including the generation of Unique QR codes for each physical book and asset that the library has since the library does not only have books, but different items as well like chess boards, previous made projects by students and such. Management of data replication and backup, this process holds control over the management of data into the main SQL server and the backup storage that will be decided by the clients. Fully automated backups is a process that we are keen on clarifying with the client if they want that option or if they want a more manual process regarding backups due to the hardware constraints of the project. Generate Management reports will generate the processes needed for the data analysis. Management of Staff Security verifies the users via password hashing and strictly enforces role-based access control for all system actions.

The Output Phase reflects the processes of all of the other two process layers. The system delivers an interactive dashboard displaying key library metrics, like the count of all of the due date books, books borrowed for the day, week, and month. The dashboard also has the transactions and the confirmations that they went through. Besides that, Structured reports are also generated such as the master inventory, the list of overdue reports, and the amount of students that have their academic quota of required books completed by that point. The dashboard will also have an indication that the system is synced with the local and main databases.

The system would be developed as a web application, using Visual Studio Code for software development, and a cloud hosting service, that is decided by the client. The system will be developed as a way to help the ease of borrowing and returning of the library inventory without the librarian being hassled with keeping the students' library cards at hand, whenever they have a book borrowed. The project will also be developed at the campus of Holy Redeemer, at their designated library.

Major Software Functions

Process Circulation

The Process Circulation subsystem constitutes the fundamental operational backbone of the library management system, handling all borrowing and returning transactions. This module contains critical functions that write data to the main server, managing the logic that oversees book circulation activities. Additionally, the Transaction Logging functionality records borrowing events by linking specific books via their QR codes to students, timestamping each transaction, and automatically setting due dates. The system simultaneously updates book status from "Available" to "Borrowed" and vice versa, ensuring real-time accuracy in inventory tracking.

Management of Inventory and Assets

The Management of Inventory and Assets component executes comprehensive CRUD (Create, Read, Update, Delete) operations for all library materials. Furthermore, this subsystem generates unique QR codes for each physical book and asset in the library's collection, accommodating not only books but diverse items such as chess boards and student-created projects. The Book Inventory Management module provides administrators with tools to maintain complete and accurate records of all library materials, supporting the addition of new books with detailed information including ISBN, author, and category classifications. The system enables updating records for new editions and removing outdated or damaged materials while organizing books by category and assigning shelf locations for efficient tracking.

Management of Data Replication and Backup

The Management of Data Replication and Backup subsystem maintains control over data synchronization between the main SQL server and backup storage solutions. This process includes fully automated backup capabilities, though the implementation approach requires client consultation regarding preferences for automated versus manual backup processes due to hardware constraints. The system ensures data integrity through scheduled automated database backups occurring at least weekly, with copies stored in both local and cloud repositories to mitigate risks related to data loss.

Generate Management Reports

The Generate Management Reports functionality compiles and organizes data from the database to create comprehensive reports based on user queries. This subsystem produces structured reports including master inventory lists, overdue book reports, transaction histories for specific students or books, and overall library usage statistics. The Monthly Library Report function provides administrators with insights into library performance through comprehensive analytics, generating reports that display borrowing trends, popular book categories, overdue items, and fine collections through visual graphs and charts.

Management of Staff Security

The Management of Staff Security subsystem verifies users through advanced password hashing techniques and strictly enforces role-based access control for all system actions. This component ensures that Admin, Librarian, and Student roles each possess appropriate permissions and access levels within the system. User Authentication validates login credentials for librarians, administrators, and students on their respective platforms, maintaining system security and data integrity through encrypted passwords and session-based authentication protocols.

QR Code Generation and Integration

The QR Code Generation component creates unique QR codes for each individual book copy added to the inventory, enabling precise tracking of specific copies borrowed by students. This functionality works in conjunction with the existing RFID system for student identification, creating a hybrid approach that leverages institutional infrastructure while enhancing individual book accountability. The system processes QR code scans during borrowing and returning transactions, capturing unique student identification codes and book QR codes to initiate circulation processes.

Dashboard and Interface Management

The system delivers an interactive dashboard displaying key library metrics including counts of due date books, books borrowed daily, weekly, and monthly. The dashboard presents transaction confirmations and provides real-time status updates, indicating whether specific books are available, borrowed, or overdue. Moreover, the interface includes indication systems showing synchronization status between local and main databases, ensuring operational transparency for library staff.

Mobile Library Access Integration

The Mobile Library Access function ensures students can manage their library accounts through mobile-optimized browsers, allowing users to log in, view borrowed books, renew items, search for new materials, and receive due date notifications directly from their mobile devices. This functionality extends system accessibility beyond the physical library location, accommodating the diverse schedules and

technological preferences of the student body while maintaining comprehensive account management capabilities.

Project Estimates

Historical Data for Estimates

The estimates for the Library Management System of Holy Redeemer School of Cabuyao were derived from previous student-led system development projects conducted within the institution. These projects, including attendance trackers and enrollment systems, were completed within 12 to 14 weeks using small development teams and open-source technologies. The similarity in project structure, tools, and objectives provided a reliable reference point for determining realistic time frames, workloads, and resource requirements.

Since the proposed library system includes advanced features like QR code generation, RFID integration, automated reporting, and a centralized web-based interface, additional time was allocated for feature development and testing. The team analyzed the performance of past projects and adjusted the estimates to accommodate these new modules and ensure system stability. Development will be conducted on a local server using XAMPP to support rapid prototyping and continuous testing, while database backups will be synchronized with cloud storage to maintain data security.

Based on historical project durations and team performance, the system's total development time is estimated at 14 weeks. This includes all major phases such as requirement gathering, interface design, programming, and deployment. The schedule allows flexibility for Agile sprints and iterative client feedback to ensure that the final product meets user needs while remaining technically feasible.

Estimation Techniques Used

Process-Based Estimation – This technique was used to break down the project into smaller framework activities such as planning, design, development, testing, and documentation. Each activity was further divided into tasks (e.g., database setup, user interface creation, QR code integration) and estimated individually. The total effort for each task was then summed to produce an overall estimate for the project. This method helped define realistic workloads and distribute them evenly among the developers.

- **Use-Case Estimation** – The system's main functions were represented through specific use-cases, such as "Book Borrowing," "Book Returning," "QR Code Scanning," and "Report Generation." Each use-case was analyzed based on complexity, user interaction, and data processing needs. Simpler use-cases like "User Login" required less time, while more complex ones such as "QR Code Generation" and "Inventory Management" were given higher time estimates. This method allowed the team to visualize the project structure and focus on

high-impact features.

- **Reconciling Estimates** – After obtaining estimates using both process-based and use-case methods, the team compared the results to check for inconsistencies. Differences in time or effort allocation were discussed and adjusted to reach a unified estimate. The reconciliation process ensured that both technical and functional perspectives were balanced, providing a more accurate overall forecast.

- **Estimation for Agile Projects** – Since the project follows the Agile-Scrum approach, each development sprint was estimated separately. Every sprint includes a set of user stories that are decomposed into smaller engineering tasks. Time estimates were assigned to each task based on past experience and complexity. The sum of all sprint estimates defines the total project duration. This iterative estimation process supports flexibility and allows adjustments based on user feedback after each sprint cycle.

These techniques collectively provided a comprehensive approach to predicting project effort, ensuring that the estimates were not only data-driven but also adaptable to changes during development.

Effort, Resource, Cost, and Project Duration Estimates

The following estimates are provided for the design and implementation of the Library Management System:

- **Effort:** Estimated 14 weeks of development work, totaling approximately **450–500 person-hours**, divided among three team members.

- **Resource Requirements:**

    1. **People:** 3 Developers (Team Members)

    2. **Hardware:** 1 Laptop (Librarian use), 2 Desktop Computers (Development and Testing), 1 QR Code Scanner

    3. **Software Tools:** Visual Studio Code, MySQL/XAMPP, GitHub, Canva, and QR Code Generator (All Free Versions)

- **Cost Estimates:**

    1. **Personnel:** ₱0 (Student Project)

    2. **Hardware:** ₱0 (School-owned and personal computers used)

3. **Software Licenses:** ₱0 (Open-source software)

4. **Printing and Materials:** ₱1,000 (QR labels, documentation, and report printing)

5. **Transportation and Miscellaneous:** ~₱800 (All members, live at different distances to the school.)

6. **Total Estimated Cost: ~₱1,800**

- **Project Duration:** Estimated at **14 weeks (approximately 3.5 months)**, divided into four major phases:

  1. **Planning and Design:** 3 weeks – Requirement gathering, system analysis, and database modeling.

  2. **System Development:** 6 weeks – Coding, module integration, and QR code functionality implementation.

  3. **Testing and Deployment:** 3 weeks – Debugging, client feedback sessions, and initial deployment.

  4. **Documentation and Training:** 2 weeks – Writing system manuals, user training, and project presentation.

The estimates ensure an organized workflow that fits the constraints of the school's resources while maintaining high standards for reliability, accuracy, and usability. The structured timeline supports incremental delivery under the Agile model, with testing and client validation at the end of every sprint. This ensures that the final system meets both functional requirements and the operational needs of Holy Redeemer School of Cabuyao.

Project Schedule

Work Breakdown Structure

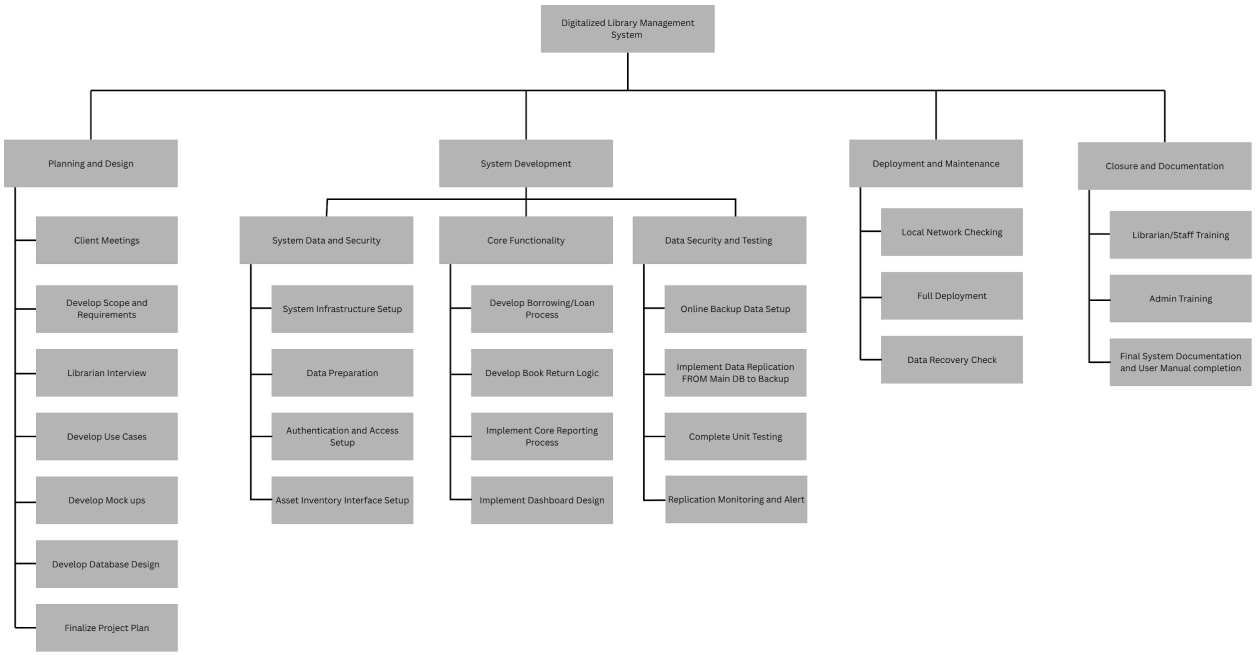
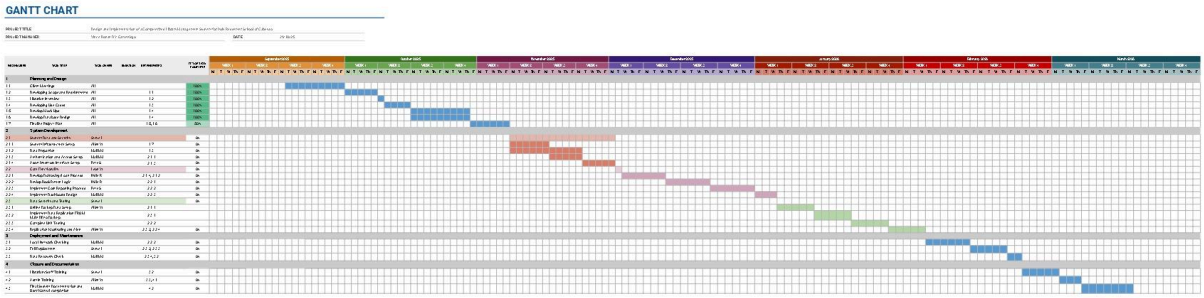


Figure : Work Breakdown Structure

The work Breakdown Structure outlines the systemic flow of the development of the project through 4 distinct phases: the Planning and Designing, System Development, Deployment and Maintenance, and the Closure and Documentation phases. The WBS is adapted to have an agile-compatible group where a column is focused on a single important aspect of the whole development of the project.

In Planning and Designing, several steps are concerned with meeting the clients about their requirements for the final output, designing preliminary options that cater to their problems, designing multiple use-cases, and then finalizing the project plan. This phase ensures that the client is onboard with what we are proposing to have the features that they want and that it is well within scope. In the System Deployment phase, the processes included in this phase are considered as individual sprints which have a Minimum Viable Product (MVP) at the end of each sprint. It is split into three main sprints, System and Data Security, Core Functionality, and Data Security and Testing. The three sprints tackle the areas of the project that deal with development databases, core functions, and the logic for connecting to the internet to make backups. Deployment and Maintenance is where the application and system as whole is implemented into the local system and unit and system testing are done. Closure and Documentation is where the user manuals, training and documentation are finalized.

Gantt Chart



Risk management

Risk Management Table

Asset	Threat	Vulnerability	Risk	Likelihood	Risk Rating
Library Devices	System Manipulation	Underpower Equipment	System crashes that lead to loss of work	High	High
Database and stored records	Unauthorized access	Lack of encryption	Compromises Student's Information	Medium	High
Network Connection	Unstable Connection	Limited Bandwidth and unreliable provider	Interruption with Librarian and student access	Medium	Medium
Qr scanner	Hardware Compatibility Issues	Inadequate driver support	Failure to scan properly, leading to delays in transactions	Low	Low
Project Timeline	Missed Deadline Due	Poor time management	Delay in delivery of key system	Medium	Medium

			modules		
Human Resources	Insufficient Training	Lack of technical Literacy	Incorrect data input	High	Medium
Software Source Code	File Corruption	Absence of backup	Loss of development process	Low	High
Physical Document	Damage or loss	No back up of paper based files	Loss of reference for manual record	Low	Low

Mitigation and Prevention Strategies

1. Regular Backups:
   Implement scheduled automated database backups at least once per week to mitigate risks related to data loss. Copies of backups will be stored in both local and cloud repositories.

2. Version Control System (VCS):
   Utilize GitHub as the repository for source code management, allowing for version tracking, rollback capability, and collaborative editing without overwriting work.

3. Hardware Maintenance:
   Ensure regular maintenance of existing library computers. Allocate resources for the potential replacement of malfunctioning hardware units.

4. System Optimization and Testing:
   Perform regular unit, integration, and performance testing to identify inefficiencies before deployment.

5. User Training:
   Conduct training workshops for librarians, students, and administrative staff to improve proficiency in using the system and avoid operational misuse.

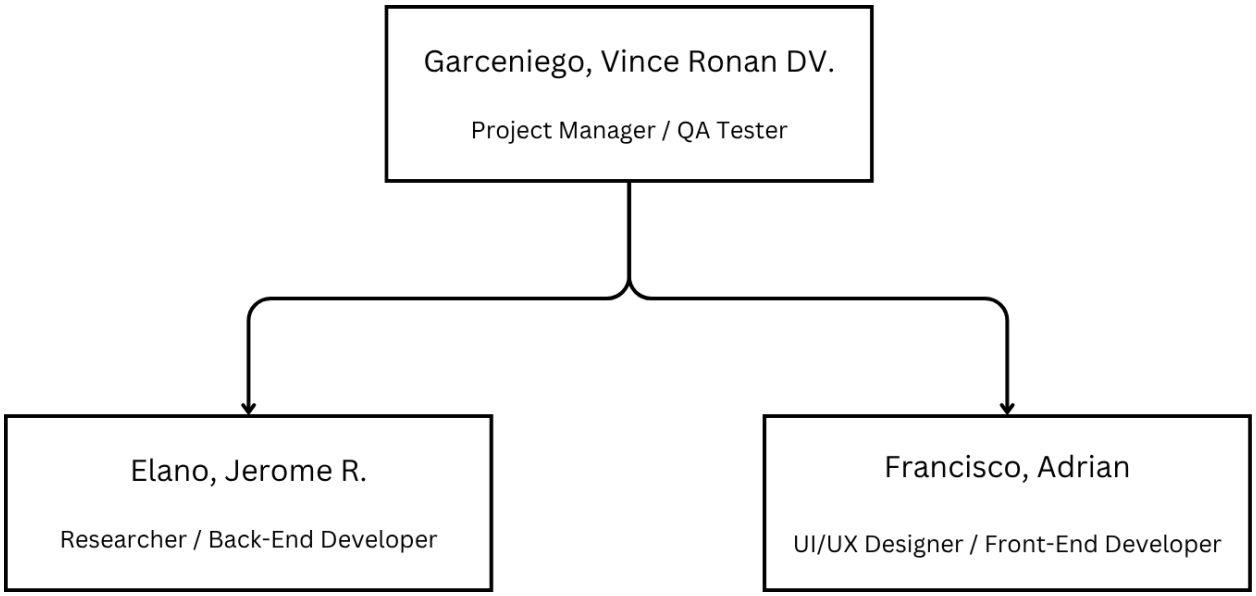
6. Enhanced Security Protocols:
   Integrate strong encryption, hashed passwords, and session-based authentication. Employ access-level permissions for Admin, Librarian, and Student roles.

7. Internet Redundancy:
   Coordinate with the school's IT staff to ensure a reliable network connection or backup internet plan during system use.

8. Documentation and Reporting:
   Maintain system documentation logs and update a “Lessons Learned” file after each sprint or testing phase to continuously refine mitigation measures.

Staff Organization

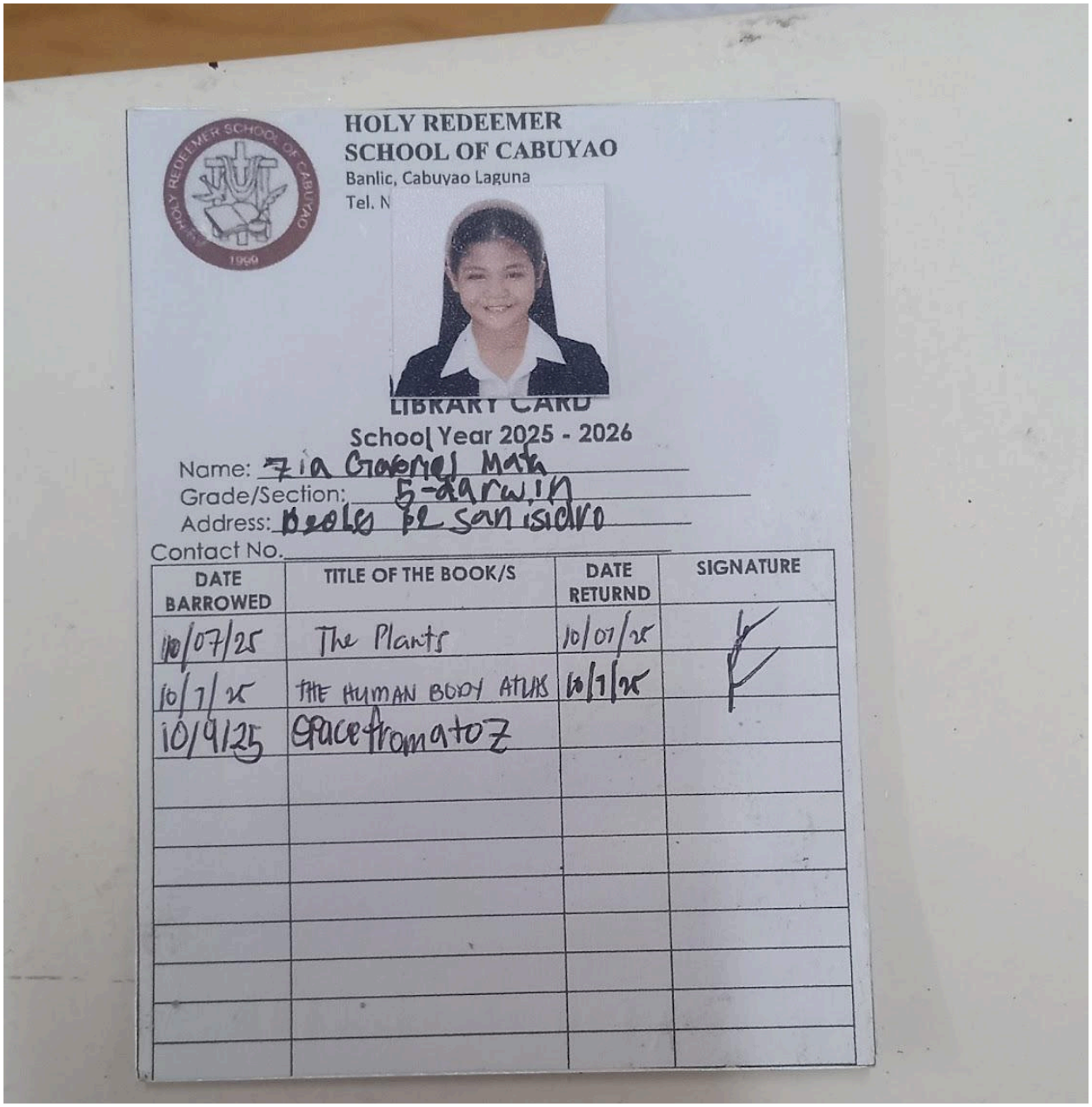
Team Structure



Management Reporting and Communication

1. Daily - Asynchronous: Messenger used for primary communication, Discord for Calls with sharing screens.
2. A checklist: a checklist in Discord as well as a copy in a shared Google Drive for all documents.
3. Github & Git: for version control of all codes.
4. A Weekly Meeting

Appendix:



HOLY REDEEMER
SCHOOL OF CABUYAO
Banlic, Cabuyao Laguna
Tel. N

LIBRARY CARD
School Year 2025 - 2026

Name: Zia Gabriel Mark
Grade/Section: 5-darwin
Address: Beales pl San isidro
Contact No.

DATE BARROWED	TITLE OF THE BOOK/S	DATE RETURND	SIGNATURE
10/07/25	The Plants	10/07/25	
10/7/25	THE HUMAN BODY ATLAS	10/7/25	
10/9/25	Grace from a to Z		

Sample Student Library Card

Accession	Date Acquired	Author	Title of Book	Edition	Copyright	Pages	Source of Fund	Cost Price	Publisher	Place	Remarks
000001		Brando B. Bagaoisi	Making Job Hunting Fun		2017	184			Vibal Group Inc.	2017	
000002		Leonora Ana E. Bautista	Learning and Living in the Economic and Political Dimensions		2017	578			Rex Bookstore Inc.	2017	
000003		Ana B. Vanluna	Effective Home Economics and Livelihood Skills in HELE		2017	340			Adriatico Pub. Co.	2017	
000004		Ma. Virginia Bancoro	Handbook for Young Housekeepers	Revised		304			Goodwill Trading Co. Inc.	2016	
000005		Esther C. Cayanan	HELE for Young Housekeepers	K-12		304			Rex Bookstore Inc.	2016	
000006		Romina E. Alvior	The World of Work			210			Jimpaz Pub. Inc.	2016	
000007		Cecelia M. Garcia	HELE for Life			184			Rex Bookstore Inc.	2015	
000008		Evangeline C. Bando	Shaping Life Through HELE			360			Jimpaz Pub. Inc.	2015	
000009		Anita P. Cacoc	HELE at Work			314			Vibal Group Inc.	2014	
000010		Alyn L. Bonito	Learn, Earn and Progress			241			Armel Printing	2013	
000011		Arsenio D. Bayao	The World of Work			122			Nelson Book Store Inc.	2008	
000012		Romina E. Alvior	Learning and Living in the Economic and Political Dimensions			371			Phil. Rex Bookstore Inc.	2006	
000013		Evangeline Gonzalez	Home Economics and Livelihood Education			318			Rex Book Store	2007	
000014		Dr. Cristina N. Gonzales	Learn, Do, and Earn			355			Rex Book Store	2007	
000015		Marcelo M. Agnagan	HELE			212			Phil. Rex Bookstore Inc.	2006	
000016		Hazel P. Cayacco	Our Work of MAPÚA			212			Dr. E.C. Pub. Publishing House Inc.	2011	
000017		Teresita B. Agustin	Making Ach Meaningful			276			Modern Trading Co. Inc.	2009	
000018		Teresita B. Agustin	Growing Up with Good Manners			198			Vibal Group Inc.	2014	
000019		Lina M. Bolinao	Eng. Skills, Attitude and Study Habits			192			Vibal Group Inc.	2012	
000020		Lynn A. Maloncon	Explore and Enjoy Through Reading	Fourth		419			Alvior Pub. Inc.	2006	
000021		Beverly C. Ramos	Enjoying Good Friendship			198			Primer Book Pub. Inc.	2006	
000022		Lea B. Jimenez	Keeping In Touch w/ Math			214			Vibal Group Inc.	2013	
000023		Norma B. Sajdrella	Skills to Effective Communication			217			Goodwill Trading Inc.	2011	
000024		Teresita B. Agustin	Moving Up with English 6			244			The Rex Bookstore Inc.	2006	
000025		Nilda Ann C. de Jesus	English Interactions 6			320			Phoenix Publishing House	2009	
000026		Lourdes M. Bilog	Across Frontiers through Reading			444			Phoenix Publishing House	2014	
000027		Raquel C. Olpoc	Enjoying Reading 6	Second		340			Rex Book Store	2011	
000028		Alicia R. Bambico	Rainbows in English 6			514			Abiva Publishing House	2014	
000029		Verne B. Cosme	Practical English Rules by Grades 6			374			Rex Book Store	2012	
000030		Ema O. Gerhing	Essential English 6			307			D.E.C. Publishing House	2016	
000031		Virginia S. Viernes	Communications in Reading 6			314			Rex Book Store	2011	
000032		Pia Abigail Pinkoy	English Encounters: Reading 6			357			Rex Book Store	2012	

Physical Records of Inventory

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